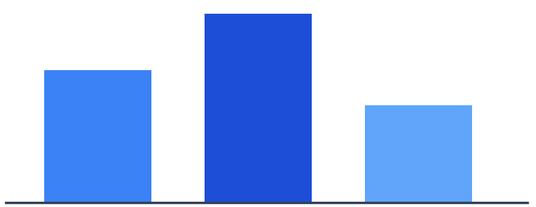
Fitness Feedback

Swarm Evaluation

	f_1	f_2	f_3	…	f_D
$\mathbf{p}_1$	1	0	1	0 1 0 …	1 0 0
$\mathbf{p}_2$	0	1	0	1 1 0 …	0 1 1
$\mathbf{p}_3$	1	1	0	0 1 0 …	1 0 0
⋮					
$\mathbf{p}_N$	0	1	1	0 0 1 …	1 1 0

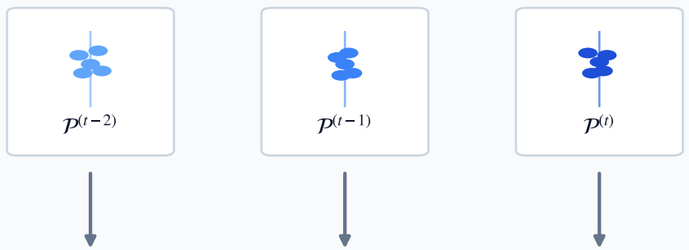


Fitness Evaluation

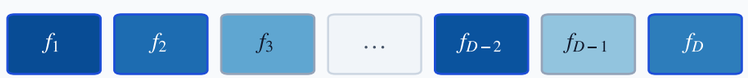
$$f(\mathbf{x}, \mathbf{w}) = w_{\text{acc}} \cdot \text{Error}(\mathbf{x}) + w_{\text{feat}} \cdot \frac{\|\mathbf{x}\|_0}{D}$$

Knowledge Accumulation

Historical Multi-Generation Experience



Cumulative Feature-Importance Matrix W

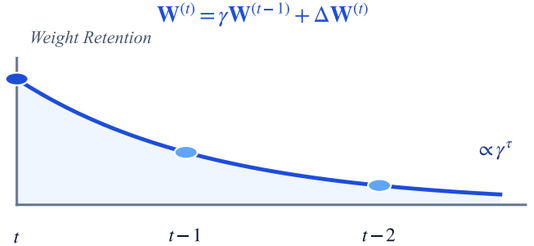


High W_j: Key Features

Low W_j: Redundant Features

Dynamic Memory Decay

$$\mathbf{W}^{(t)} = \gamma \mathbf{W}^{(t-1)} + \Delta \mathbf{W}^{(t)}$$



Knowledge-Guided Elite Refinement

Elite Solution $\mathbf{x}_e$ (Before Refinement)

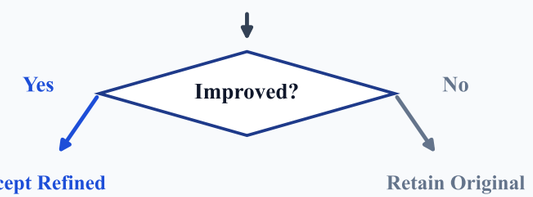
0	1	1	0	1	0	1	…	0	1

Knowledge-Guided Directional Flip

1	1	1	0	**0**	0	1	…	0	1

$0 \to 1$ $\quad$ $1 \to 0$

Fitness Re-evaluation



Elite Solution $\mathbf{x}_e^{\star}$ (After In-Place Selection)

1	1	1	0	**0**	0	1	…	0	1

Closed-Loop Evolutionary Knowledge Feedback Channel